IOMED

IOMED.HEALTH

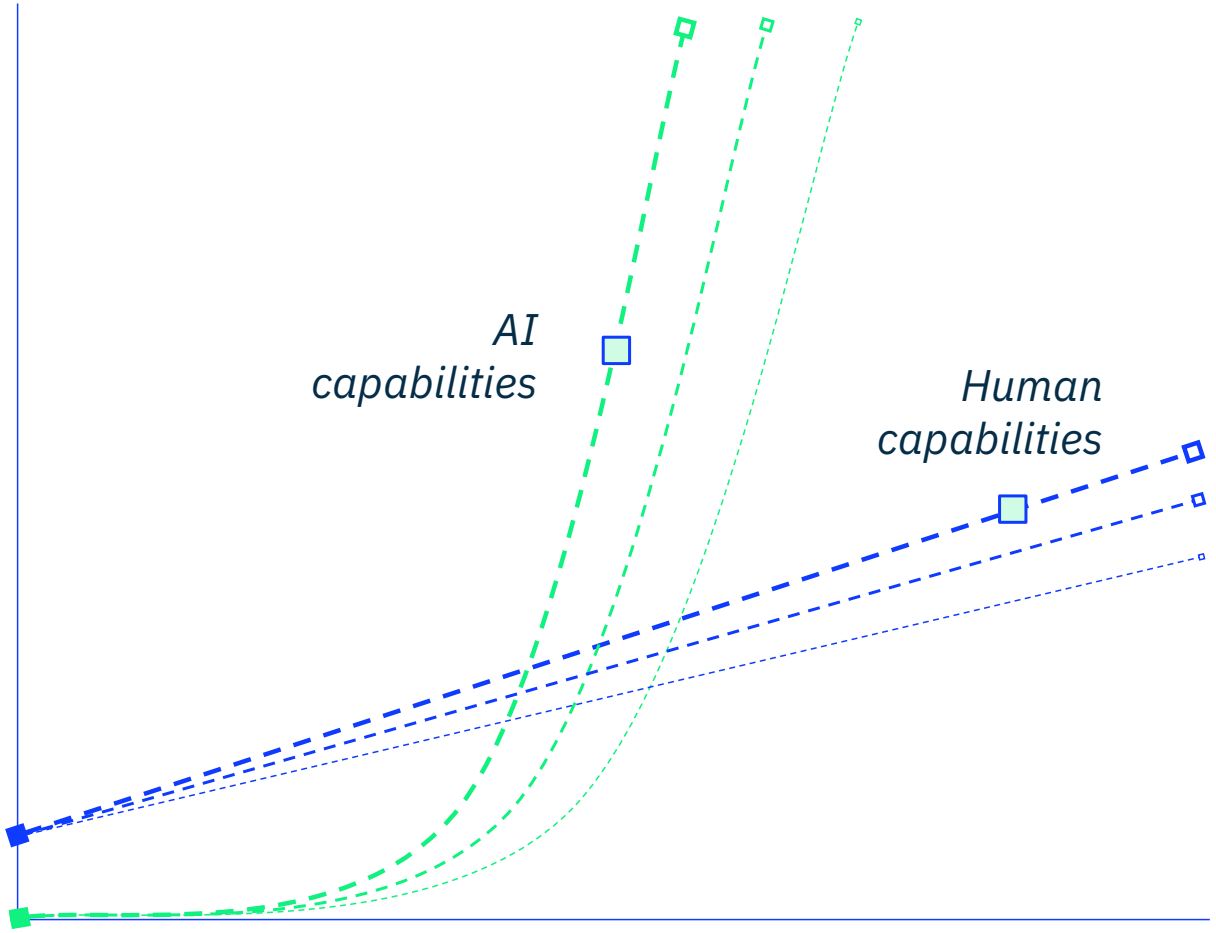
Rol de Inteligencia Artificial en estudios con datos de la vida real

Gabriel Maeztu M.D. - Director Médico y Técnico @ IOMED

IOMED

La IA como es el músculo que nos permite alcanzar una gran escala

No se trata de replicar capacidades humanas, sino más bien de una respuesta autónoma a un problema específico que necesita solución a gran escala.



IOMED

→ Donde?

1. Mapeo y **Estandarización**
2. **Análisis** de los datos
3. Optimización de **Procesos**

IOMED

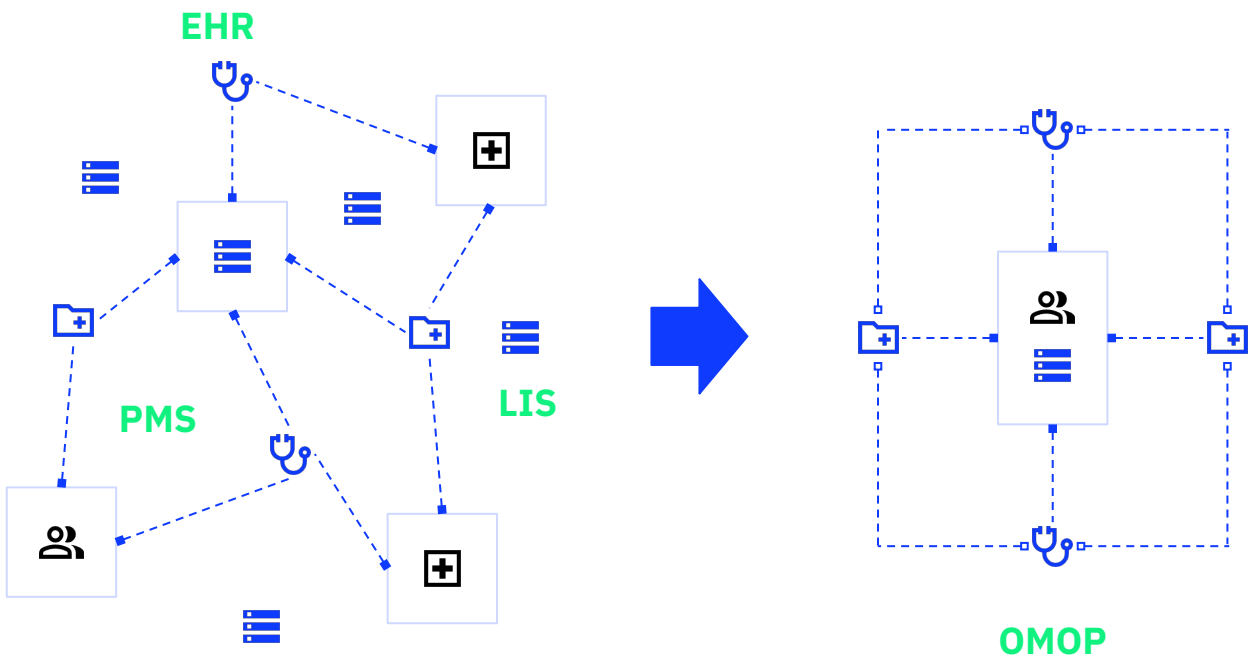
→ Usos de la IA

1. Mapeo y Estandarización

- El objetivo principal es **estandarizar** los datos del centro a gran escala.
- Esto permite acceder a nuevas fuentes de datos como:
  - Textos clínicos
  - Imagen médica
  - Datos ómicos

Creando datos paciente-céntricos

- La **digitalización** en los hospitales se ha **centrado en los procesos del hospital** y no en representar pacientes.
- En cada centro tiende a haber **varios sistemas de información** que dan soporte a procesos diferentes:
  - Historia Clínica Electrónica
  - Sistemas de gestión de Laboratorio
  - Farmacia y dispensación
  - Sistema de gestión de visitas
  - [...]
- El proceso de mapear los datos al OHDSI OMOP CDM se trata de construir **una versión digital del paciente** sumando los datos de múltiples fuentes, convirtiendo datos proceso-céntricos en datos paciente-céntricos.



Operational Data Architecture

Patient-centric Data Architecture

IOMED

1. Extracción
2. Estandarización
3. Calidad

OHDSI Tools

White Rabbit

Rabbit In a Hat

Equipo responsable

ACHILLES

DQD

Expertos en los datos evalúan las fuentes y definen la transformación de los datos al OMOP CDM

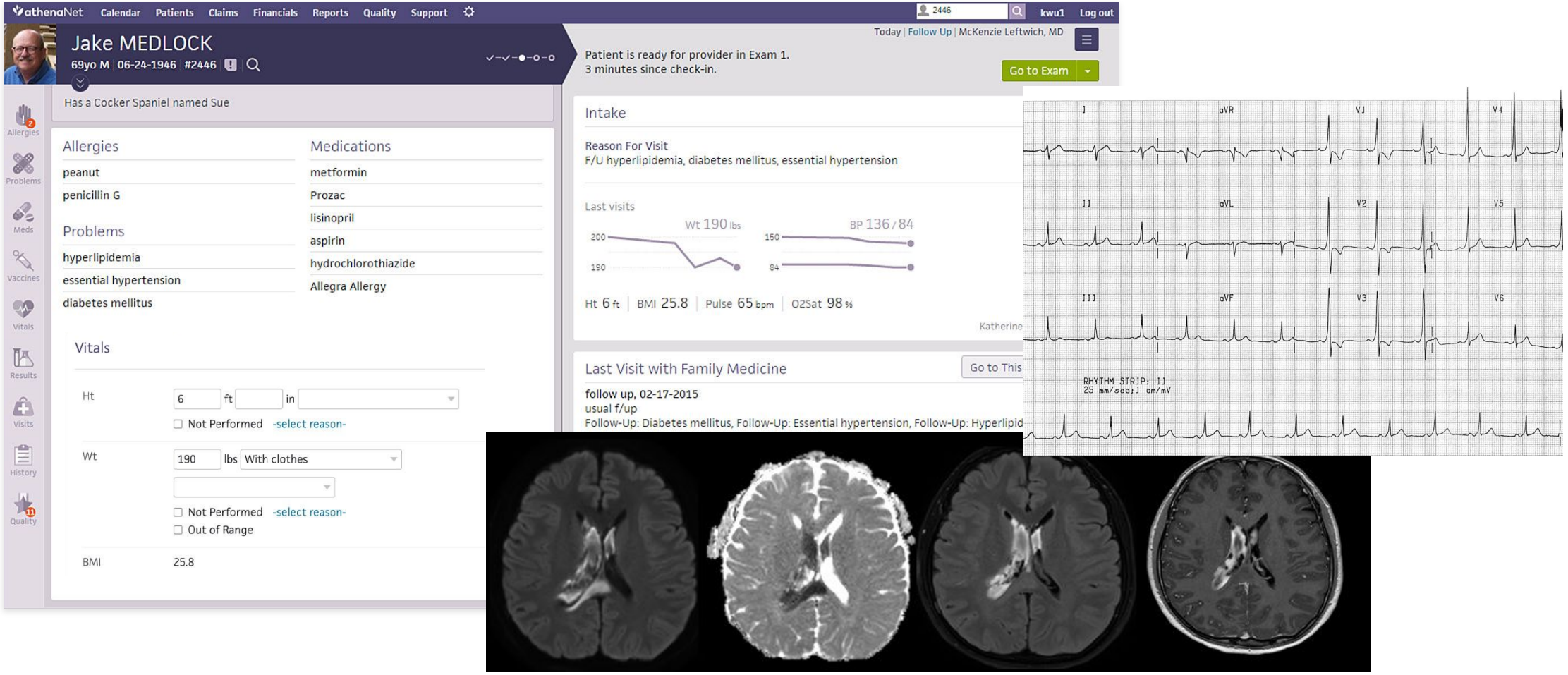
Personas con conocimientos médicos crean las asignaciones de códigos.

La calidad de los datos es evaluado, y se replantean los procesos de carga de datos y mapeos.

4. Ajustar

IOMED

→ Estructurando datos a gran escala

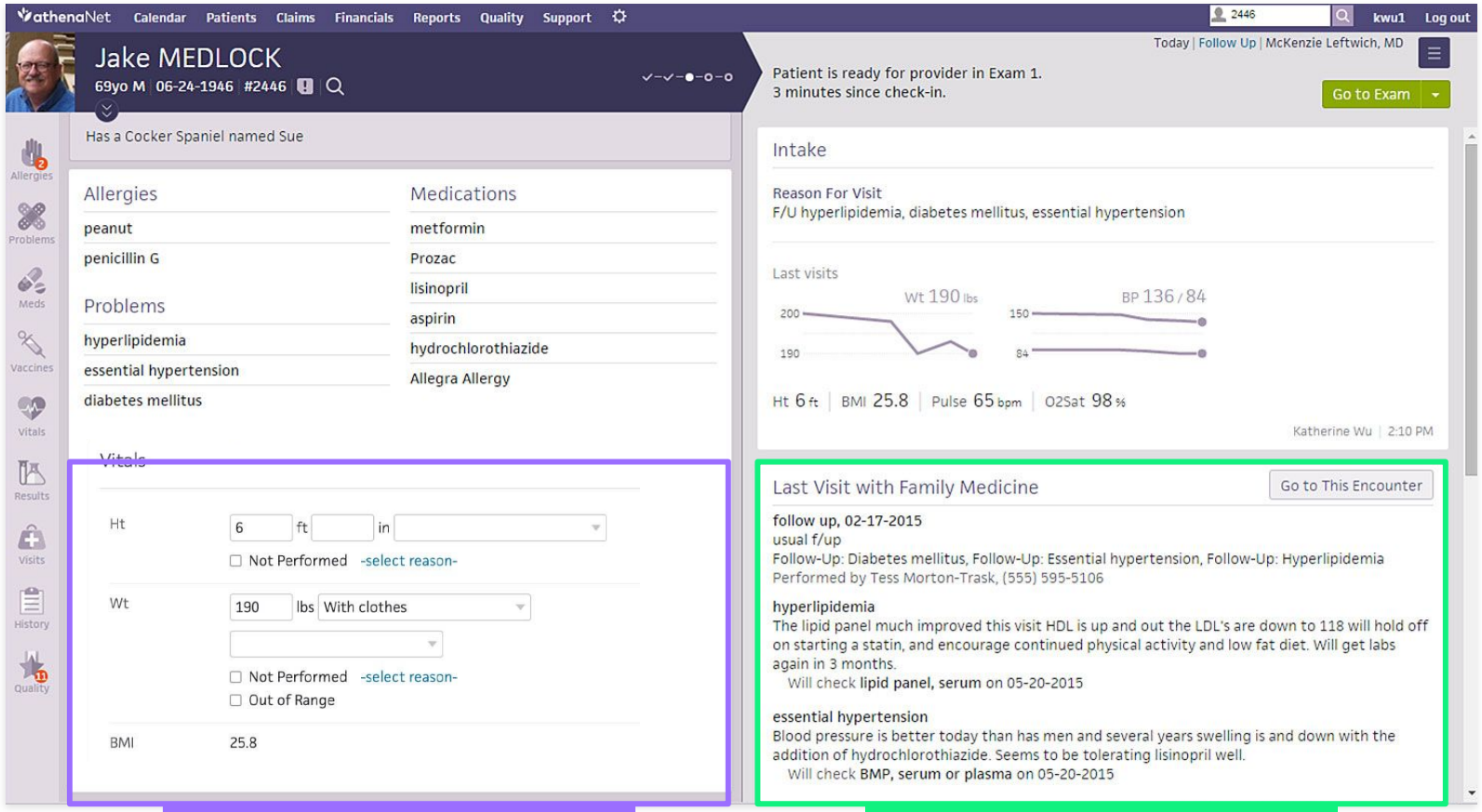


IOMED

Estructurando datos a gran escala

IOMED

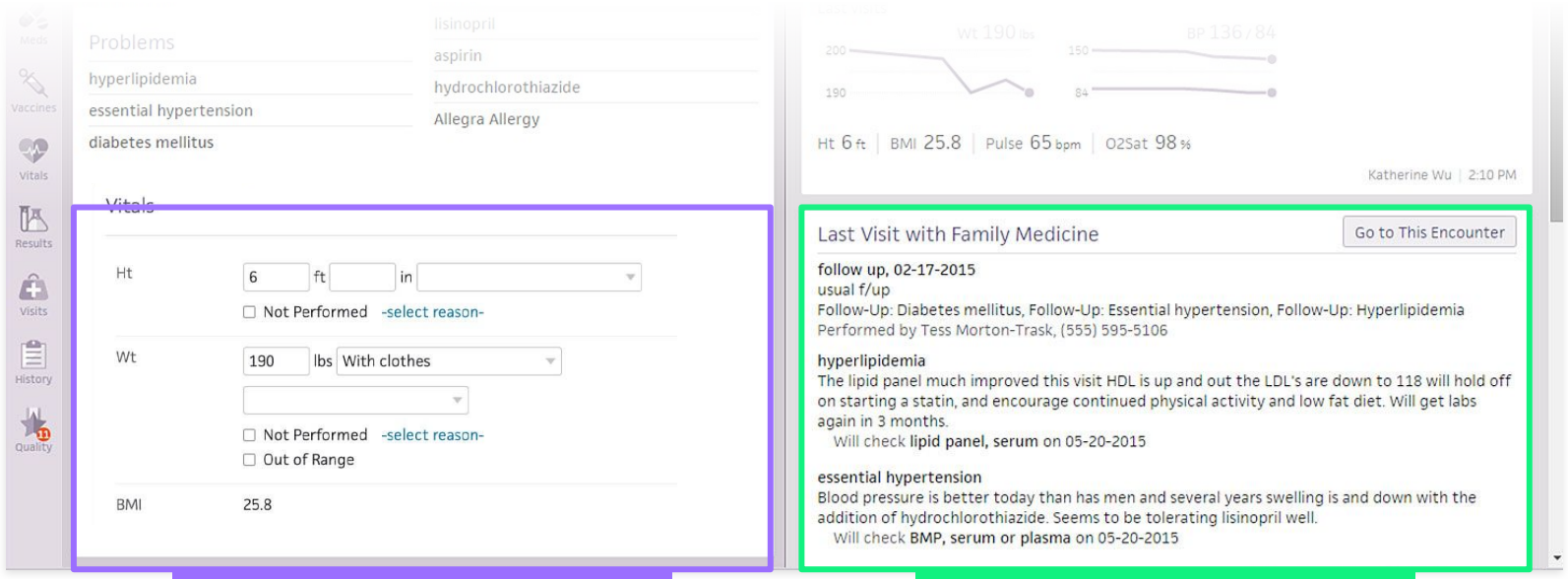
→ Estructurando datos a gran escala



STRUCTURED DATA
20% of the patient information

CLINICAL NOTES
80% of the patient information

→ Estructurando datos a gran escala



STRUCTURED DATA
20% of the patient information

CLINICAL NOTES
80% of the patient information

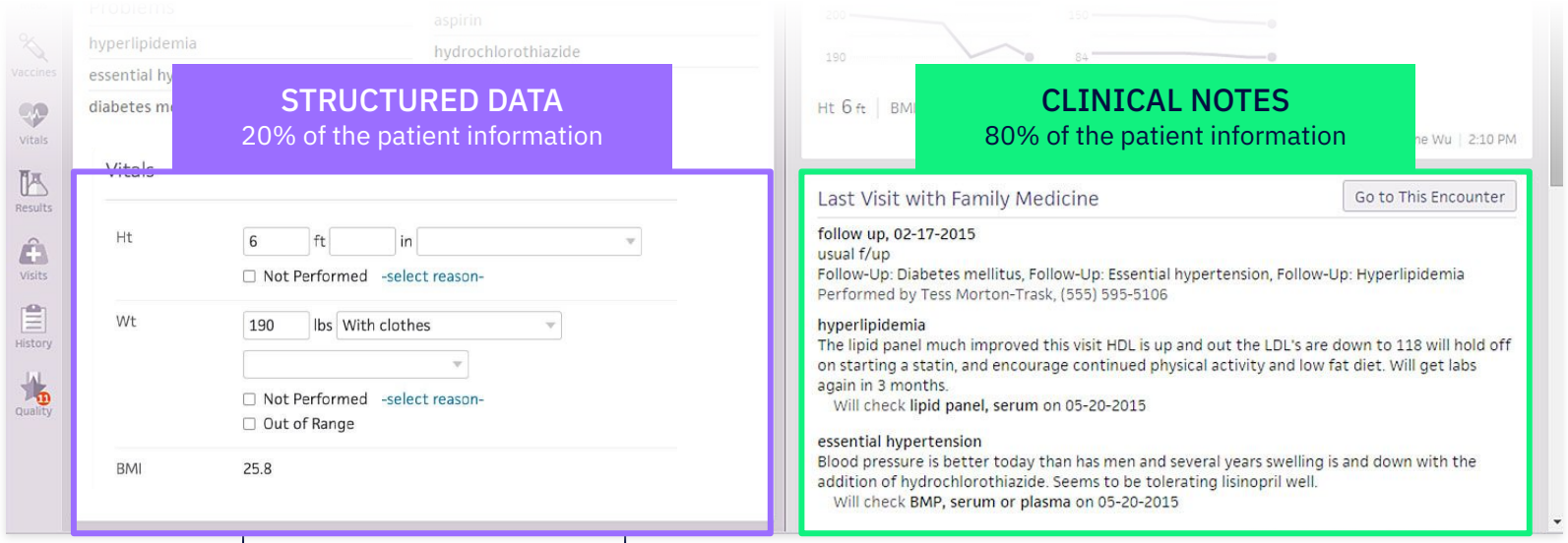
ID	patient_id	data_id	date	value	unit	visit
643	243	weight	2021-10-12	190	lbs	367
2891	243	weight	2021-11-21	170	lbs	458

noteID	person_id	date	value	visit
1	243	2021-10-12	follow up, 02-17-2015 usual f/up Follow-Up: Consultation on pian left arm since yesterday. Thi smorning pain persists. Had similar pain whn had AMI. Has taken nitrostat without improvement. Request of ECG	367

Non-standard codification

Data stored in free-text format

Estructurando datos a gran escala



...	person_id	measurement_concept_id	measurement_date	value_as_number	...
...	243	3025315	2021-10-12	86.18	...

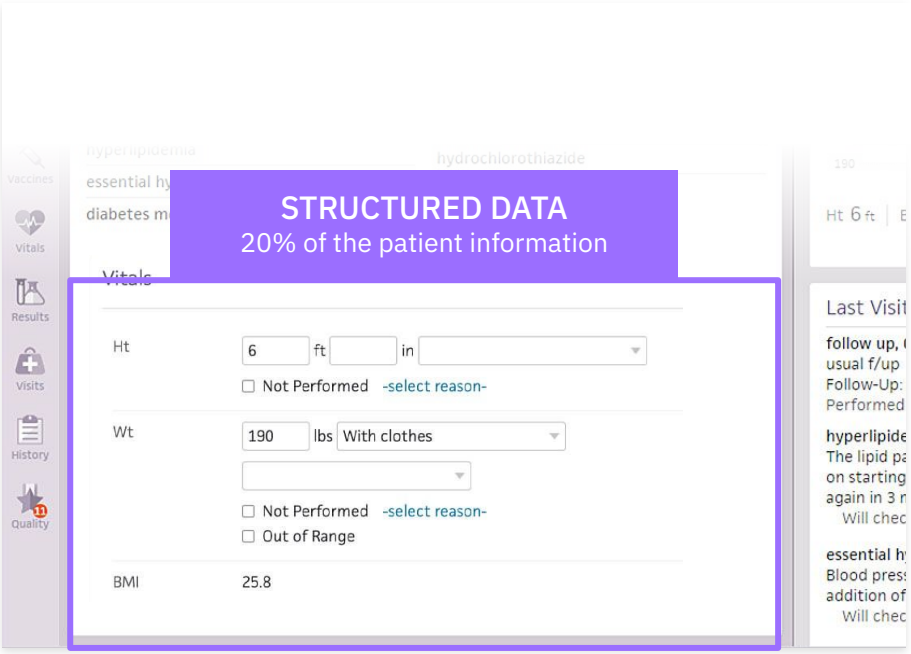
Weight → OMOP CDM Measurement

...	person_id	condition_concept_id	condition_date	visit_occurrence_id	...
	243	4115169	2021-10-12	896	

Pain in the left arm → OMOP CDM Condition Occurrence

IOMED

[01] Automated Terminology Mapping



Non-standard codification

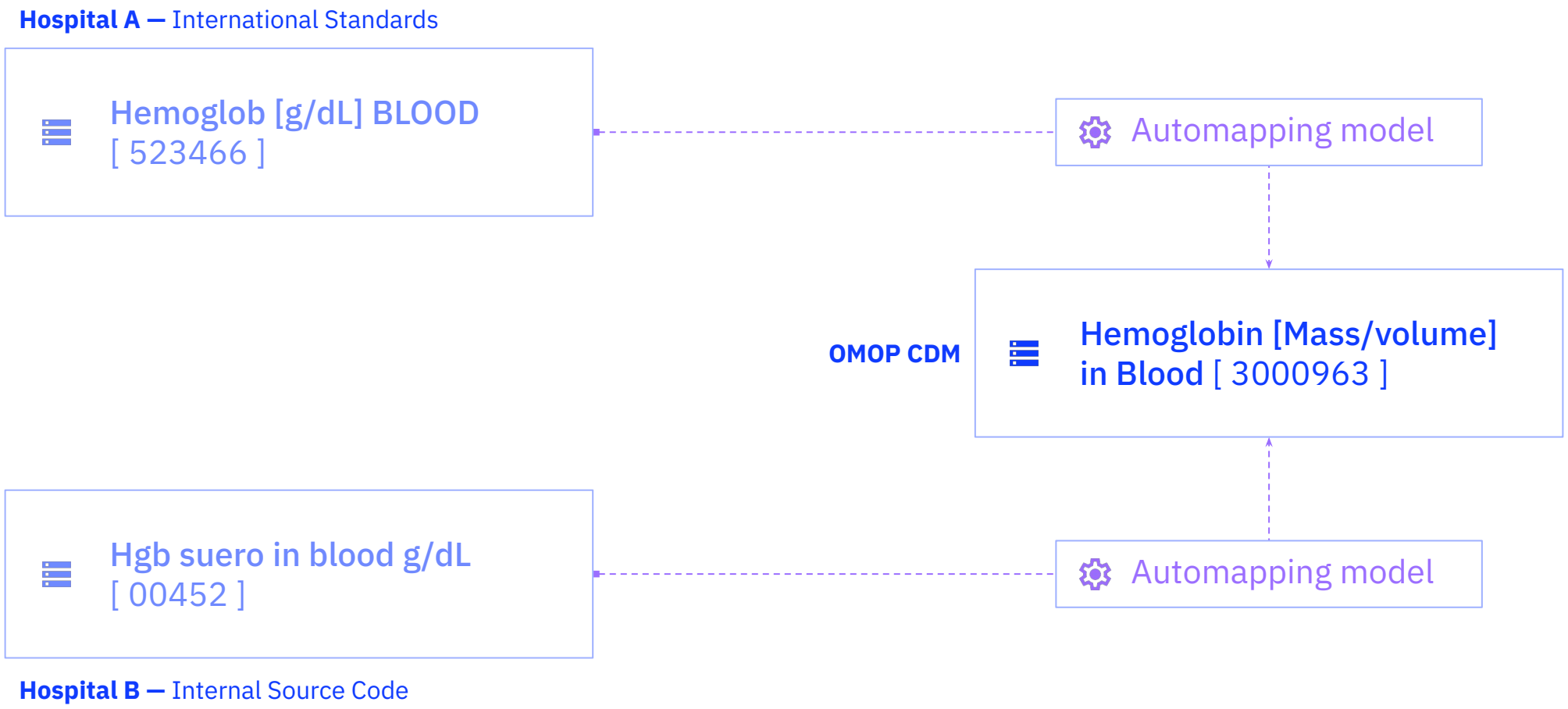
ID	patient_id	data_id	date	value	unit	visit
643	243	weight	2021-10-12	190	lbs	367
2891	243	weight	2021-11-21	170	lbs	458

Automapping model

...	person_id	measurement_concept_id	measurement_date	value_as_number	...
...	243	3025315	2021-10-12	86.18	...

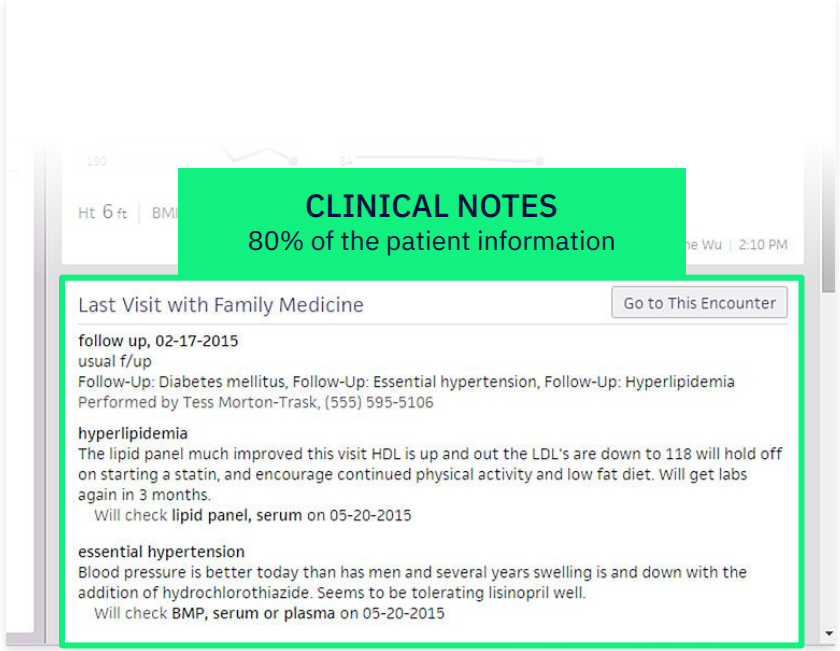
OMOP CDM Measurement

[01] Automated Terminology Mapping



IOMED

[02] Natural Language Processing



Data stored in free-text format

noteID	person_id	date	value	visit
1	243	2021-10-12	follow up, 02-17-2015 usual f/up Follow-Up: Consultation on pian left arm since yesterday. Thi smorning pain persists. Had similar pain whn had AMI. Has taken nitrostat without improvement. Request of ECG	367

Natural Language Processing

...	person_id	Condition_concept_id	condition_date	visit_occurrence_id	...
...	243	4115169	2021-10-12	896	...

OMOP CDM Condition Occurrence

[02] Natural Language Processing

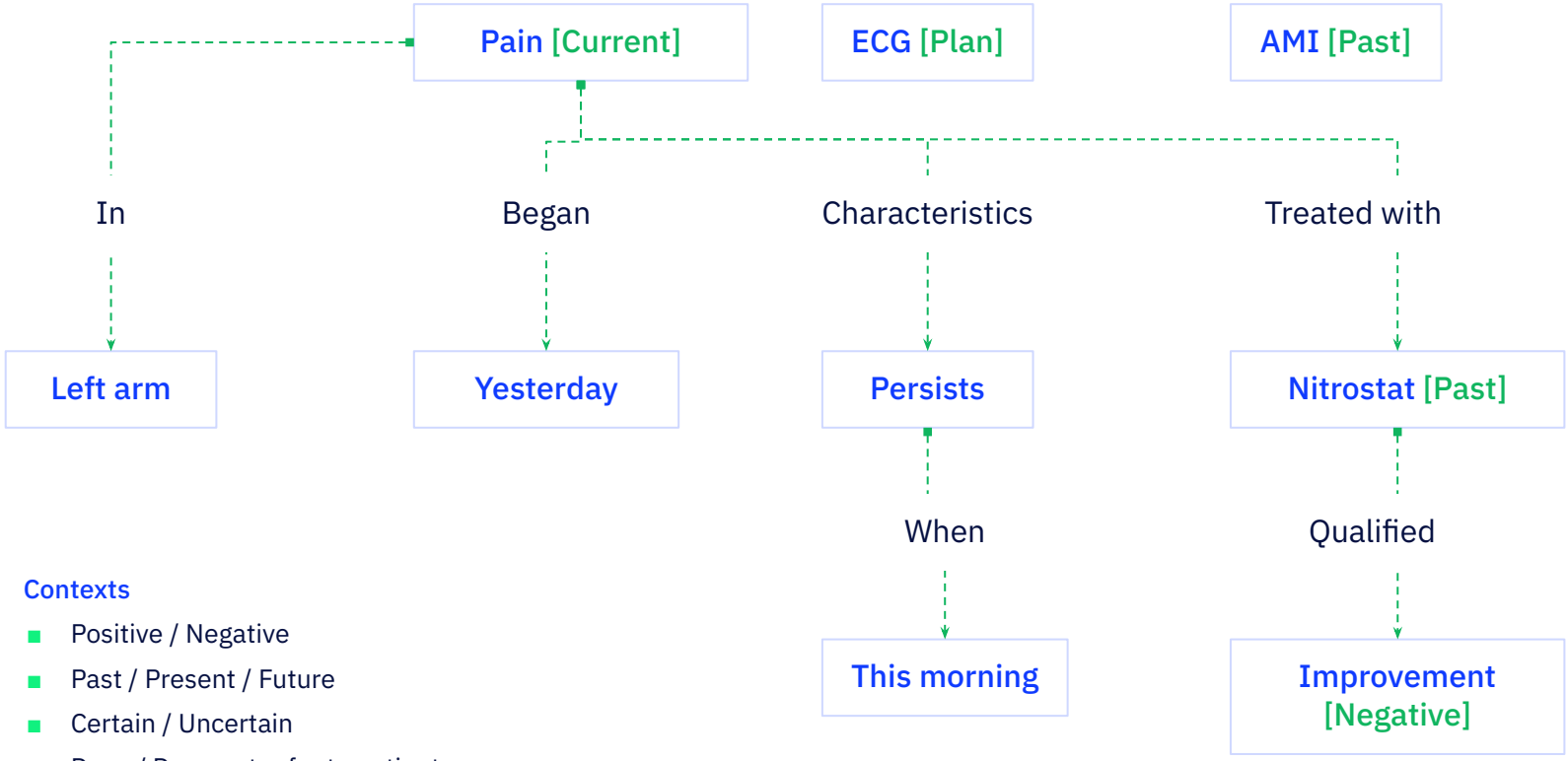
Consultation on pian left arm since yesterday.
Thi smorning pain persists.
Had similar pain whn had AMI.
Has taken nitrostat without improvement.
Request of ECG.

Natural Language Processing

- Pain [4329041]
- ECG [36308175]
- This morning -
- Improvement [3223949]
- AMI [312327]
- Left arm [4283159]
- Nitrostat [40071533]
- Persists [4038788]
- Yesterday -

[02] Natural Language Processing

Consultation on pian left arm since yesterday.
Thi smorning pain persists.
Had similar pain whn had AMI.
Has taken nitrostat without improvement.
Request of ECG.



Contexts

- Positive / Negative
- Past / Present / Future
- Certain / Uncertain
- Does / Does not refer to patient
- ...

[02] Natural Language Processing

Consultation on pian left arm since yesterday.
Thi smorning pain persists.
Had similar pain whn had AMI.
Has taken nitrostat without improvement.
Request of ECG.

Pain [Current, Certain, Patient...] — 4329041

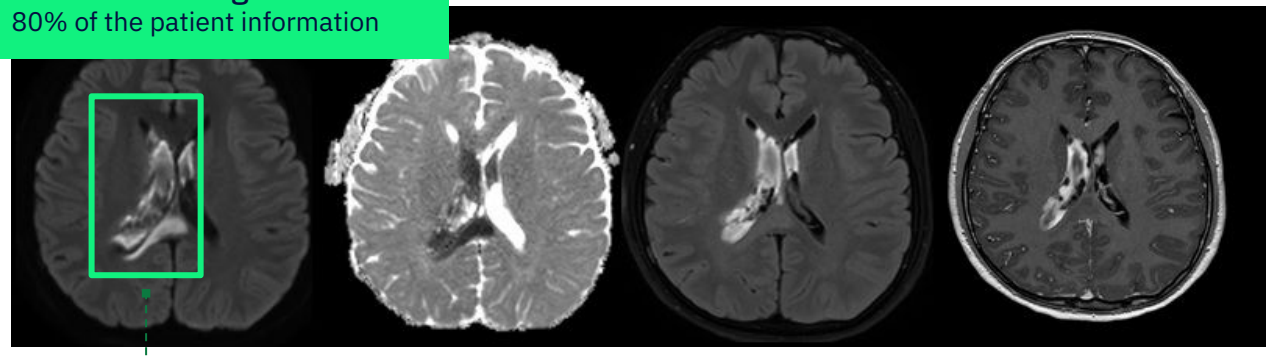
In

Left arm [Current, Certain, Patient...] — 4283159

PAIN IN LEFT ARM	
Domain ID	Condition
Concept Class ID	Clinical Finding
Vocabulary ID	SNOMED
Concept ID	4115169
Concept code	287045000
Validity	Valid
Concept	Standard
Synonyms	Pain in left arm (finding) Pain in left upper limb
Valid start	31-Jan-2002
Valid end	31-Dec-2099

[03] Computer Vision

MRI images
80% of the patient information

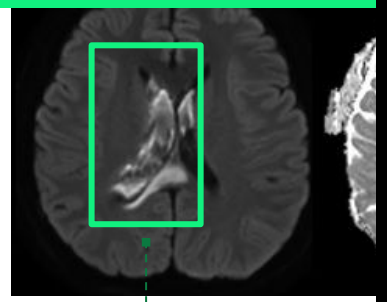


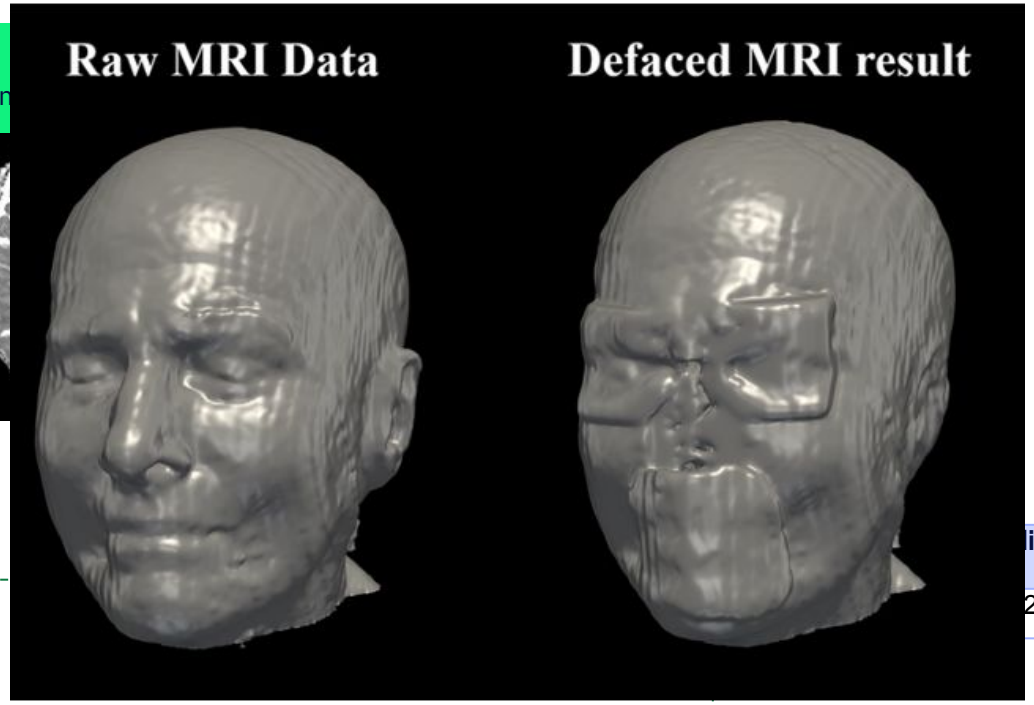
...	person_id	condition_ concept_id	condition_date	visit_occurrence_id	...
	243	36684840	2021-10-12	896	

Acute Stroke → OMOP CDM Condition Occurrence

[03] Computer Vision

MRI images
80% of the patient information





ition_date	visit_occurrence_id	...
21-10-12	896	

Acute Stroke → OMOP CDM Condition Occurrence

[04] Aseguramiento de la calidad

El Aseguramiento de la Calidad es un factor clave para garantizar la precisión, fiabilidad y rendimiento de un estudio.

La calidad de la IA está directamente relacionada con la calidad de los datos que ofrecerá el estudio que ejecutemos.

Verificación de datos
Revisión al detalle

Validación de datos
Revisión de tendencias

[04] Aseguramiento de la calidad

Verificación de datos
Revisión al detalle

El objetivo es evaluar, a partir de una muestra, la corrección del trabajo realizado para asegurar la necesaria confianza y trazabilidad.

True Positives (TP)
Patients identified by our algorithms and corroborated by a human annotator.

False Positives (FP)
Patients identified by our algorithms but discarded by a human annotator.

False Negatives (FN)
Patients discarded by our algorithms but identified by a human annotator.

True Negatives (TN)
Patients discarded by our algorithms and discarded as well by a human annotator.

Paciente varón de 56 años originario de Málaga. Diagnosticado de adenocarcinoma de próstata oligometastásica con M1 ilíaca derecha irradiada en curso de bloqueo hormonal. Anatomía patológica gleason 8 con afectacón perineural. En radioterapia paliativa antiálgica en columna lumbar iniciada el 03.21. Acude para valoración y seguimiento. Inicio Eligard mensual marzo 2021, se pasó a semestral el 29/08/2021 + Bicalutamida continuo. Actualmente solo con eligard por intolerancia a la bicalutamida.

EF: BEG, abdomen blando y depresible no doloroso a la palpación, no se palpa globo vesical. PPL negativa. Pene sin lesiones, prepucio retráctil. Testes en bolsa escrotal anodulares no dolorosos. TR: no doloroso, próstata G I, simétrica de consistencia fibroelástica de bordes bien delimitados sin palparse nódulos.

IOMED

[04] Aseguramiento de la calidad

Verificación de datos
Revisión al detalle

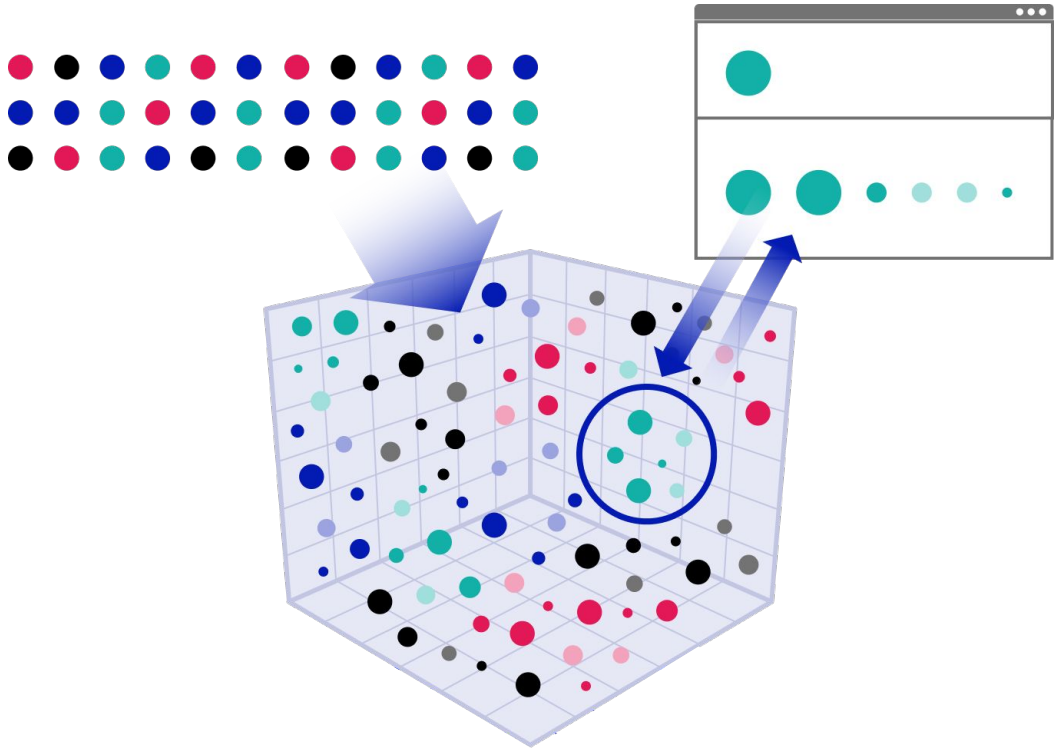
El objetivo es evaluar, a partir de una muestra, la corrección del trabajo realizado para asegurar la necesaria confianza y trazabilidad.

True Positives (TP)
Patients identified by our algorithms and corroborated by a human annotator.

False Positives (FP)
Patients identified by our algorithms but discarded by a human annotator.

False Negatives (FN)
Patients discarded by our algorithms but identified by a human annotator.

True Negatives (TN)
Patients discarded by our algorithms and discarded as well by a human annotator.



[1]: Quijada, M., Vivó, M., Abella-Bascarán, Á., Chocrón, P., Maeztu, G.d. (2022). A Framework for False Negative Detection in NER/NEL. Natural Language Processing and Information Systems. NLDB 2022 https://doi.org/10.1007/978-3-031-08473-7_30

[04] Aseguramiento de la calidad

Verificación de datos
Revisión al detalle

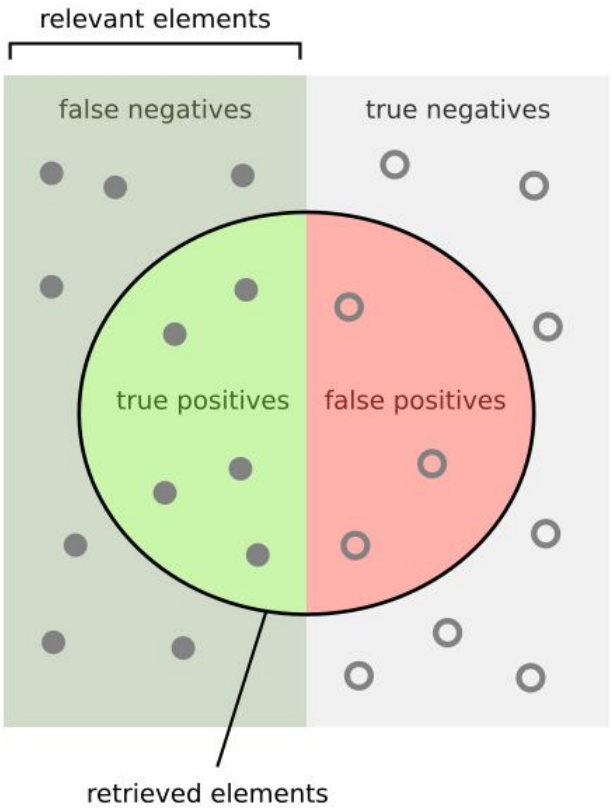
El objetivo es evaluar, a partir de una muestra, la corrección del trabajo realizado para asegurar la necesaria confianza y trazabilidad.

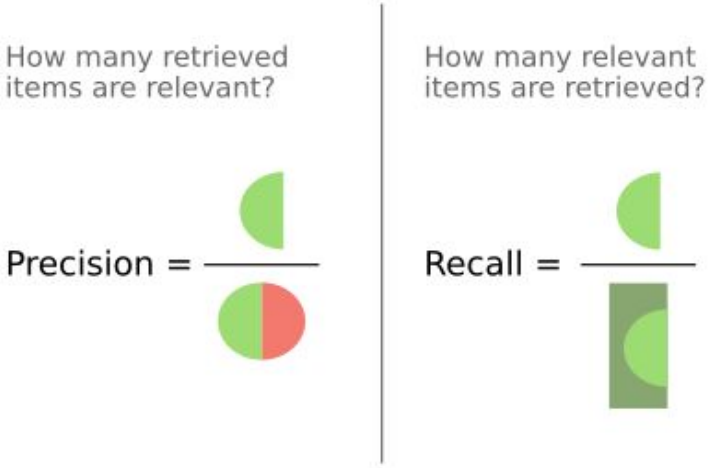
True Positives (TP)
Patients identified by our algorithms and corroborated by a human annotator.

False Positives (FP)
Patients identified by our algorithms but discarded by a human annotator.

False Negatives (FN)
Patients discarded by our algorithms but identified by a human annotator.

True Negatives (TN)
Patients discarded by our algorithms and discarded as well by a human annotator.





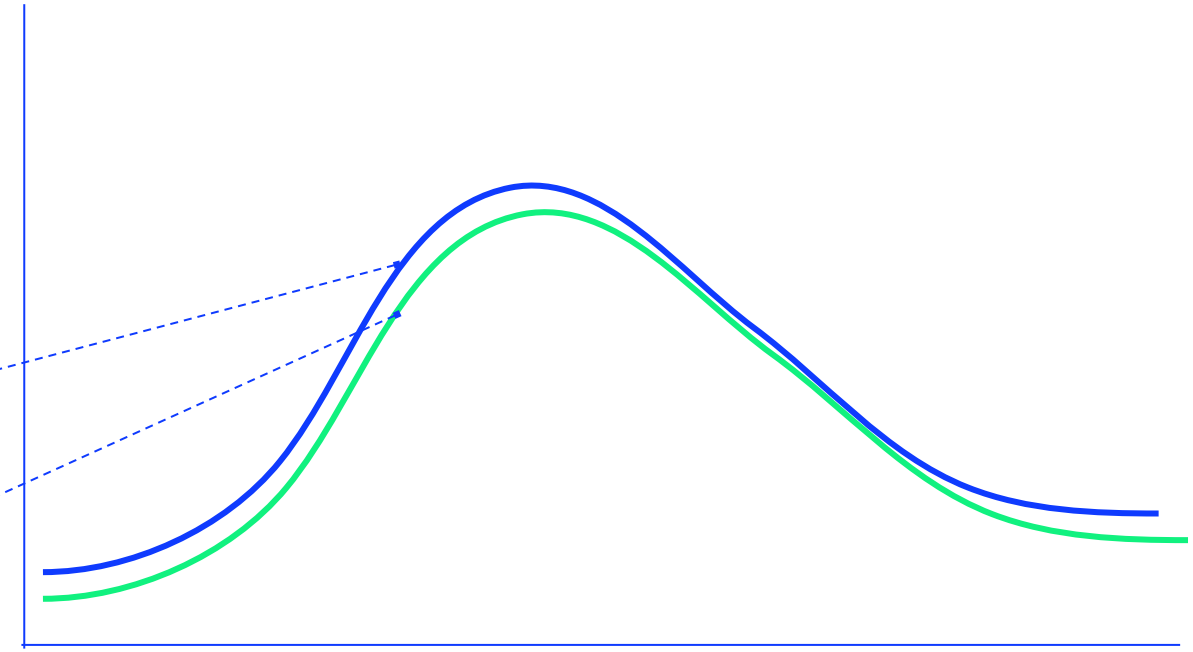
[04] Aseguramiento de la calidad

Validación de datos
Revisión de tendencias

El objetivo es evaluar todos los datos para poder comprender las tendencias y casos extremos.

Observado:
tendencia visible en los datos

Esperado:
tendencia esperada en los datos



[04] Aseguramiento de la calidad

Validación de datos
Revisión de tendencias

El objetivo es evaluar todos los datos para poder comprender las tendencias y casos extremos.

DATA QUALITY ASSESSMENT

SYNTHEA SYNTHETIC HEALTH DATABASE

DataQualityDashboard Version: 2.0.0.100
Results generated at 2022-10-12 10:45:28 in 15 mins

	Verification				Validation				Total			
	Pass	Fail	Total	% Pass	Pass	Fail	Total	% Pass	Pass	Fail	Total	% Pass
Plausibility	2179	36	2215	98%	287	0	287	100%	2466	36	2502	99%
Conformance	996	11	1007	99%	180	0	180	100%	1176	11	1187	99%
Completeness	415	33	448	93%	12	4	16	75%	427	37	464	92%
Total	3590	80	3670	98%	479	4	483	99%	4069	84	4153	**98%**

2752 out of 4069 passed checks are Not Applicable, due to empty tables or fields.
1 out of 84 failed checks are SQL errors.
Corrected pass percentage for NA and Errors: 94% (1317/1400).

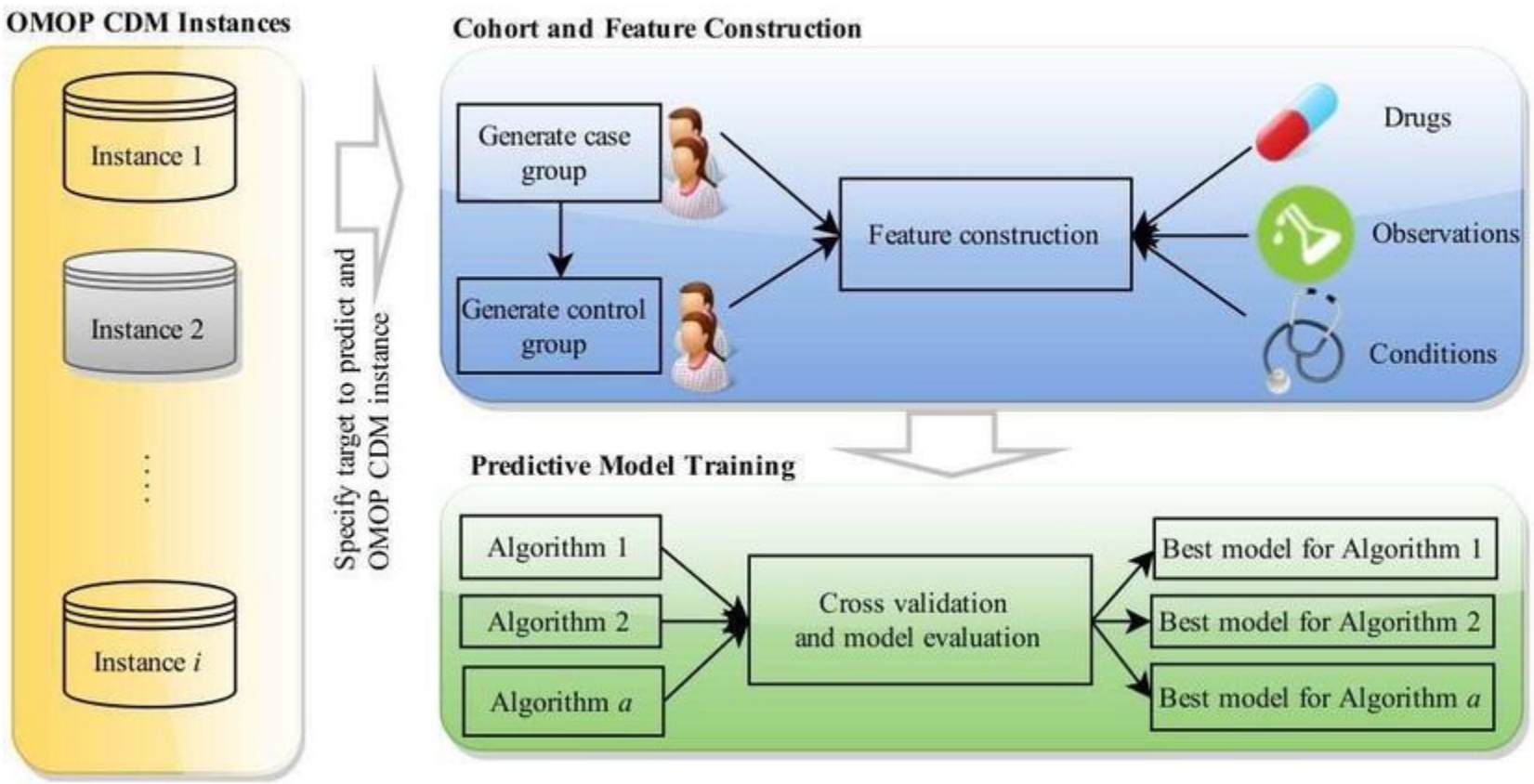
IOMED

→ Usos de la IA

2. Análisis de los datos

- El objetivo principal es **aterrizar un caso de uso** sobre los datos del centro a gran escala.
- Esto permite mejorar:
  - Conclusiones sobre las cohortes
  - Integrar análisis avanzados de forma sistemática
  - Una mejor gestión de procesos en el centro

→ Entrenamiento de modelos de Inteligencia Artificial



Rol de IA en estudios con RWD | [2] **Análisis de los datos**

IOMED

More consumer-facing domains

	Health Care Delivery Domain	Description of Application	Example of Uses of AI (Nonexhaustive)	Potential Impact on Total Mission Value (Low – Medium – High)	Current State of Adoption (Development of solutions – Pilot phase – Scaling and adaptation – Maturity)
1	Consumer	Understanding how best to engage consumers with the use of tools	Identification of patients to prioritize outreach Personalized outreach		
2	Continuity of care	Optimizing point-of-service and referrals to improve patient care	Referral integrity Patient transfers		
3	Network and market insights	Tracking relationship strength among providers	Identification of providers Benchmarking (e.g., quality)		
4	Clinical operations	Optimizing workflow of clinical operations throughout care	Hospital operations (e.g., emergency department, operating room) Capacity management Supply chain		
5	Clinical analysis	Improving patient care before, during, and after treatment	Clinical decision support Treatment recommendations Care pathway design		
6	Quality and safety	Reducing major adverse events while improving patient experience and complying with regulations	Detection of deterioration of patient's condition Regulatory compliance		
7	Value-based care	Improving performance of value-based care models	Utilization management Determination of which patients will benefit most		
8	Reimbursement	Automating and optimizing payment flows between providers and payers	Coding Prevention of denials		
9	Corporate functions	Managing back-office, administrative functions	Talent management Finance		

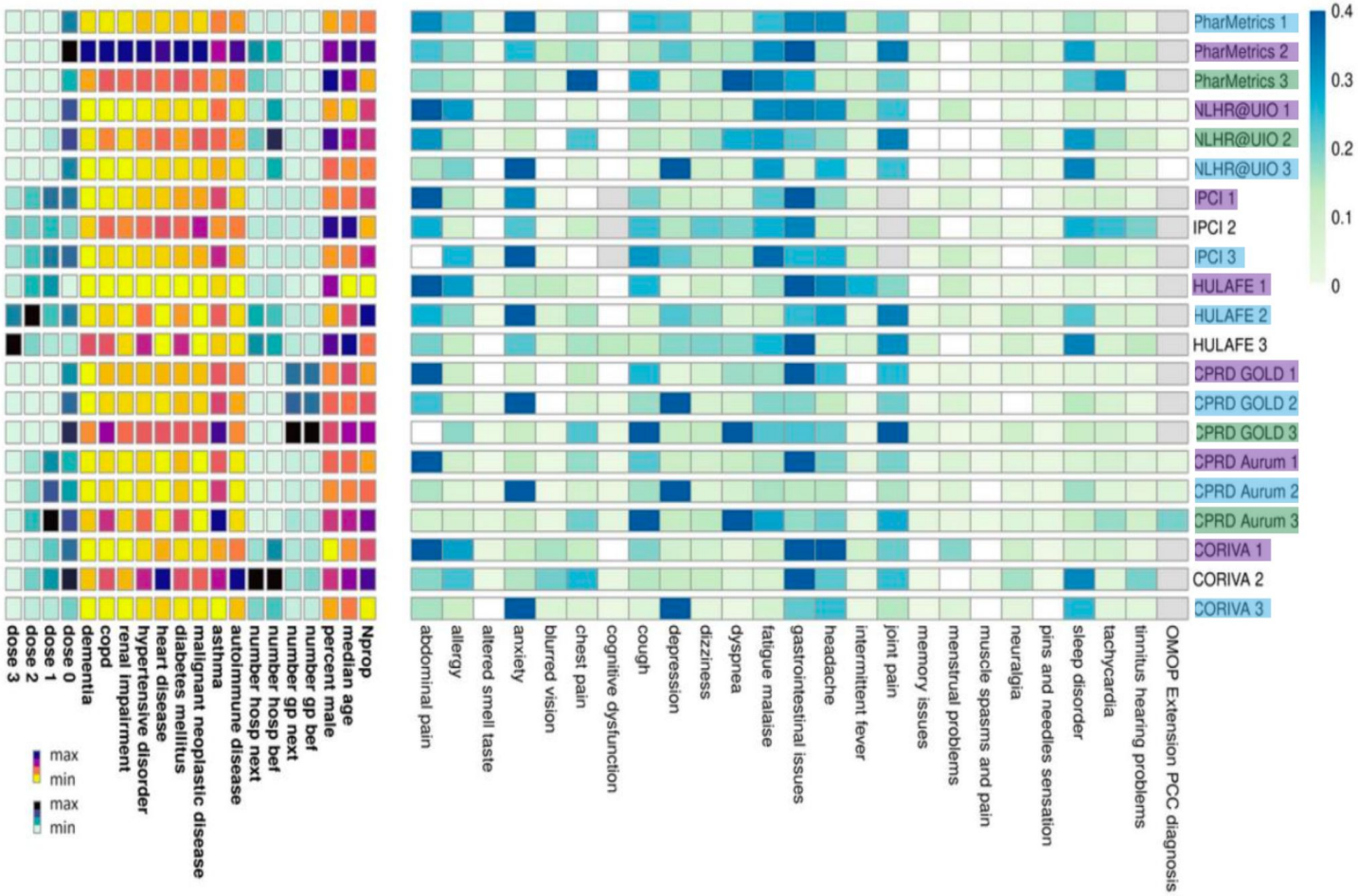
More administrative and back-office domains

N Engl J Med 2023;389:348-58. DOI: 10.1056/NEJMra2204673

IOMED

→ Clusterización de sintomatología

- Aprovechar los datos ya disponibles en el OMOP para poder hacer análisis más avanzados



Alport Syndrome

Many patients with Chronic Kidney Disease (CKD) do not have a confirmed diagnosis of Hereditary Kidney Disease (HKD) such as Alport Syndrome. **We assist physicians in identifying patients with a not filiated HKD** and supports the diagnostic confirmation process.

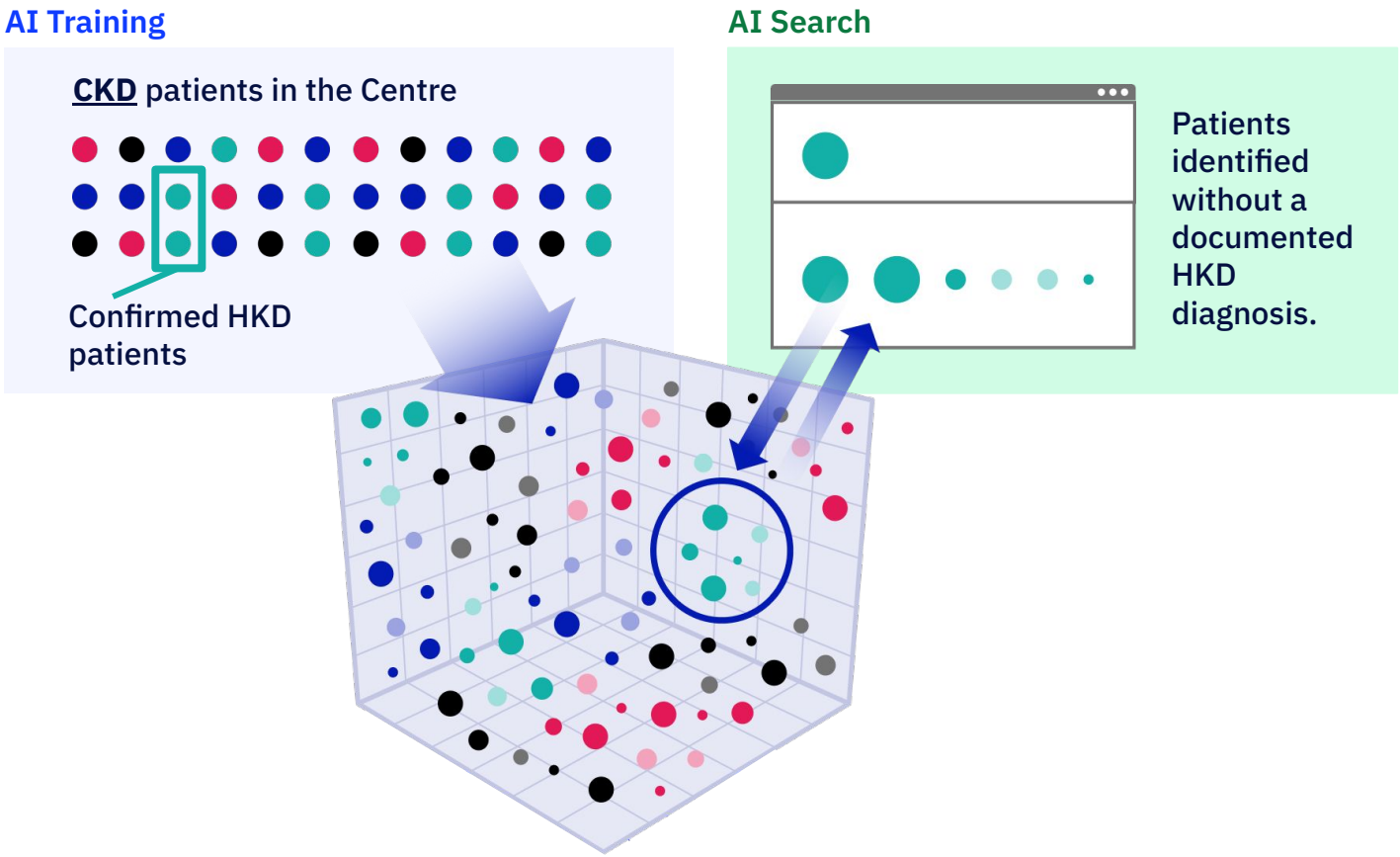
Confirmed HKD
292

New HKD
+53

CKD Patients
45.382

Patients with Chronic Kidney Disease (CKD) and a Hereditary Kidney Disease (HKD)

Hospital Fundació Puigvert (total patients=920K)



IOMED

→ Usos de la IA

3. Optimización de **Procesos**

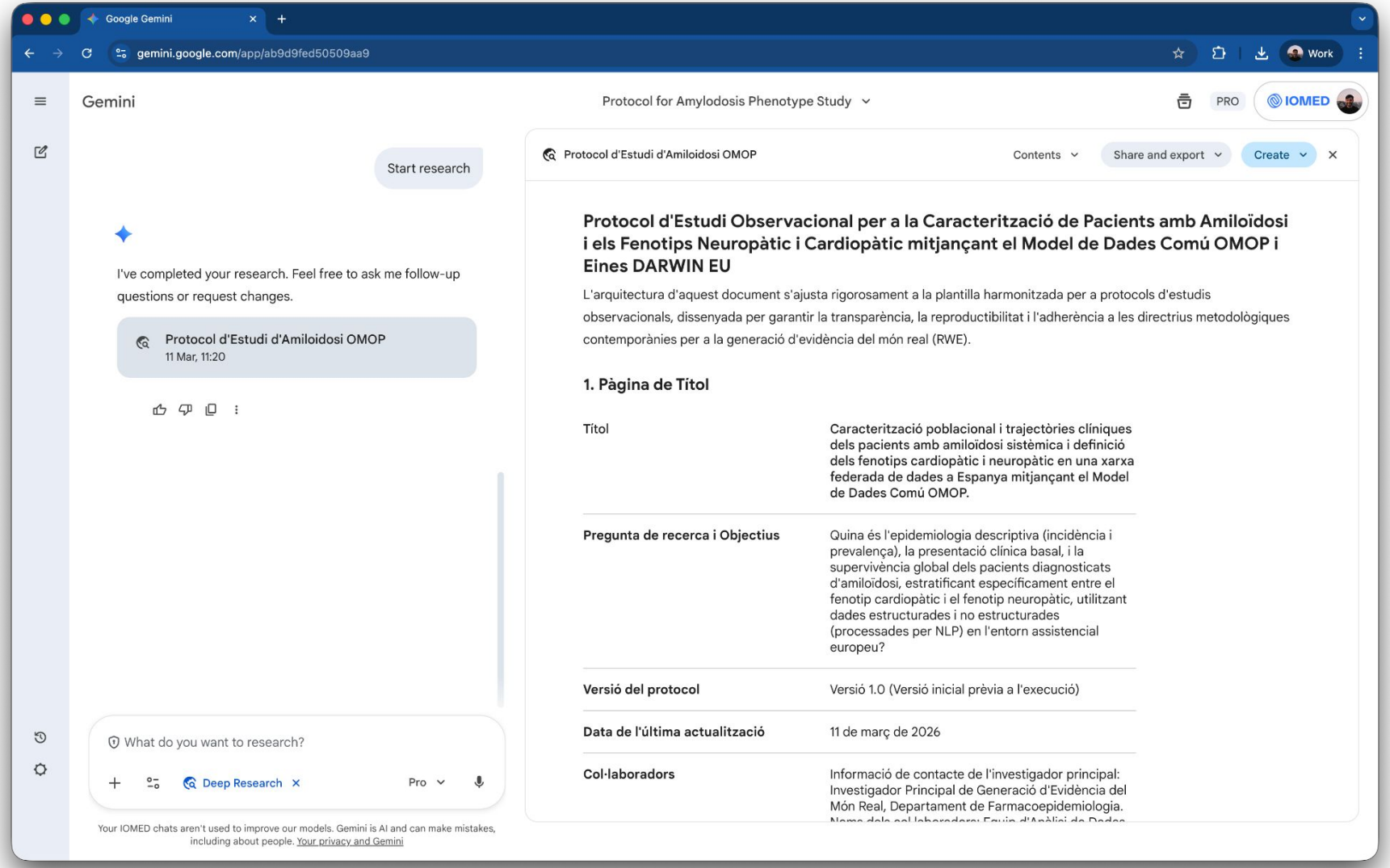
- El objetivo principal es **acelerar** los procesos necesario para ejecutar un estudio de RWD.
- Procesos como:
  - La definición del protocolo
  - La definición de los Concept Set y la Cohorte
  - El análisis de datos y la generación de resultados
  - La redacción de publicaciones

→ La definición del Protocolo del estudio

El uso de Deep Research para acelerar la búsqueda bibliográfica.

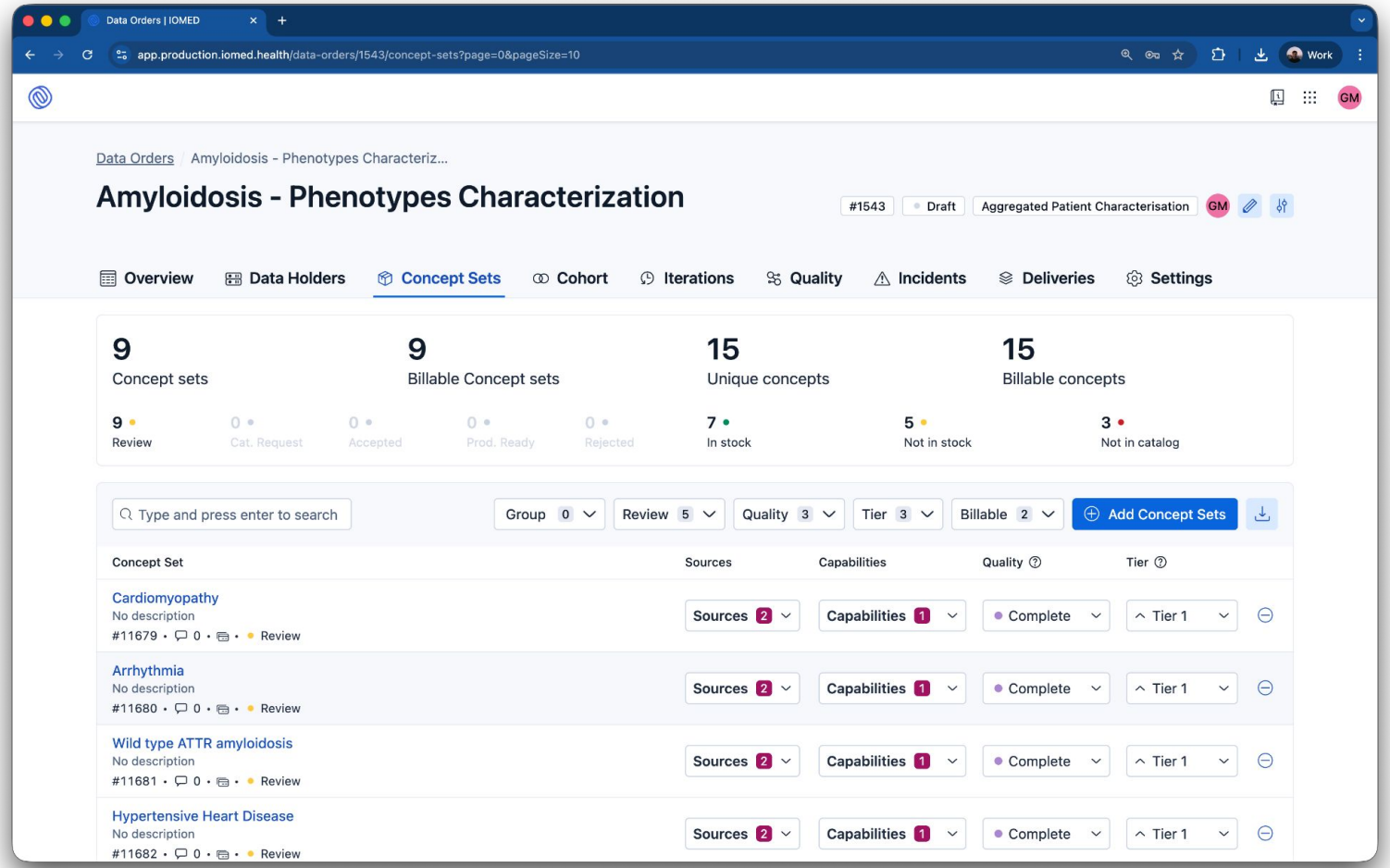
El uso de plantillas estándares para protocolos exhaustivos y de alta calidad

HARPER
HARmonized Protocol to Enhance Reproducibility



→ La definición de los Concept Sets y la Cohorte

El uso de IA para definir los ***concept_id*** del estudio de interés así como los criterios de selección.

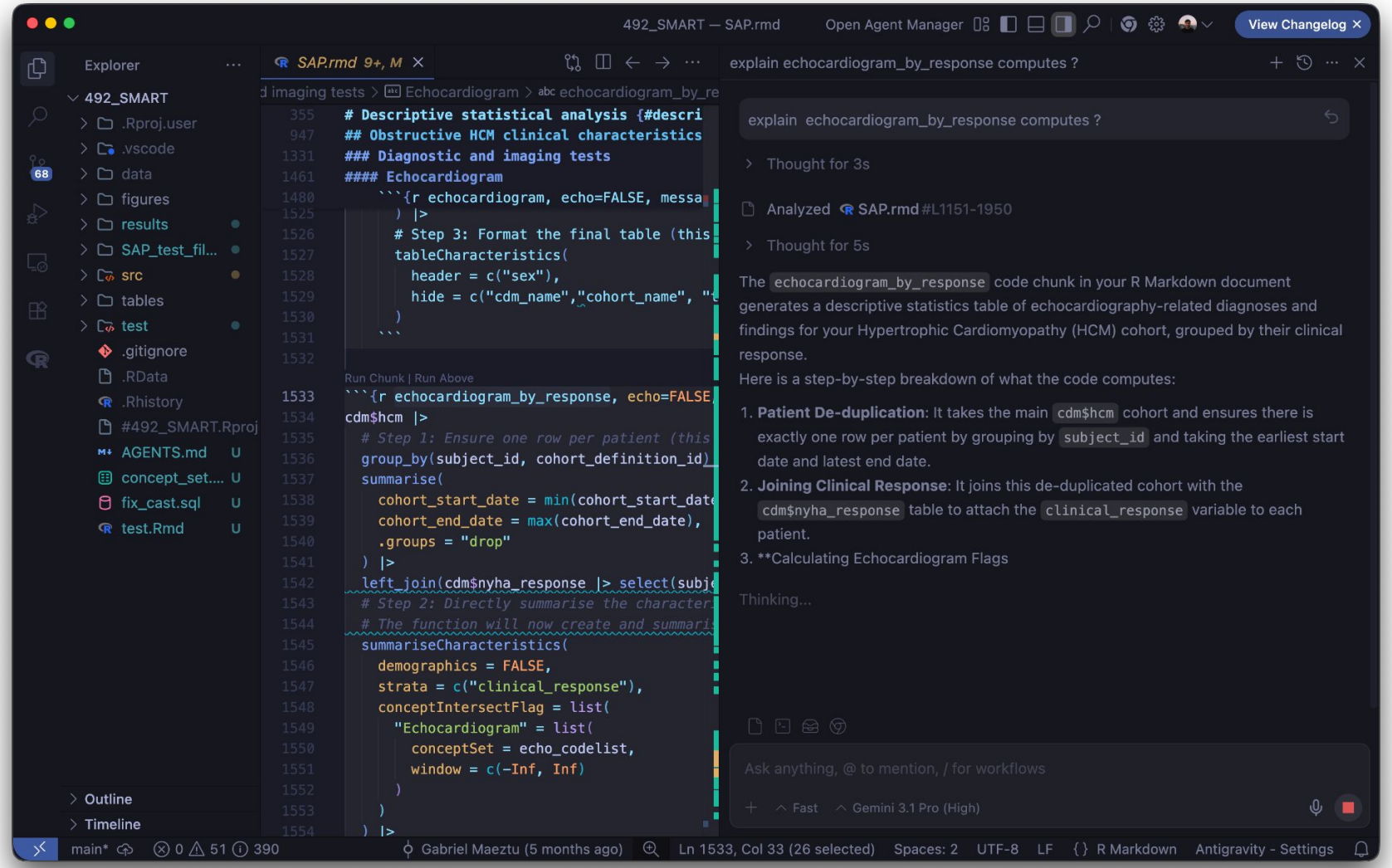


IOMED

IOMED

El análisis de datos y la generación de resultados

Generación de los scripts de R para realizar los análisis de datos y el soporte en la comprensión de las librerías de la OHDSI y la EMA.



→ La redacción de publicaciones

The NEW ENGLAND JOURNAL of MEDICINE

REVIEW ARTICLE

AI IN MEDICINE

Jeffrey M. Drazen, M.D., *Editor*, Isaac S. Kohane, M.D., Ph.D., *Guest Editor*, and Tze-Yun Leong, Ph.D., *Guest Editor*

Artificial Intelligence in U.S. Health Care Delivery

Nikhil R. Sahni, M.B.A., M.P.A.–I.D., and Brandon Carrus, M.Sc.

From the Department of Economics, Harvard University, Cambridge (N.R.S.), and the Center for U.S. Healthcare Improvement, McKinsey and Company, Boston (N.R.S., B.C.) — both in Massachusetts. Mr. Sahni can be contacted at nikhil_sahni@mckinsey.com or at the Center for U.S. Healthcare Improvement, McKinsey and Company, 280 Congress St., Ste. 1100, Boston, MA 02210.

N Engl J Med 2023;389:348-58.
DOI: 10.1056/NEJMra2204673
Copyright © 2023 Massachusetts Medical Society.

THE ADOPTION OF ARTIFICIAL INTELLIGENCE (AI) AND ITS IMPACT ON business sectors happen in phases. The use of AI is advanced in many areas, including reinventing how a financial institution provides investment advice and products to its customers, offering "recommendation engines" that suggest the next retail product to buy for a consumer who has just bought one item, and developing driverless cars. In health care delivery, however, AI remains in the early stages.

AI adoption in health care delivery lags behind the use of AI in other business sectors for multiple reasons. Early AI took root in business sectors in which large amounts of structured, quantitative data were available and the computer algorithms, which are the heart of AI, could be trained on discrete outcomes — for example, a customer looked at a product and bought it or did not buy it. Qualitative information, such as clinical notes and patients' reports, are generally harder to interpret, and multifactorial outcomes associated with clinical decision making make algorithm training more difficult. Another challenge is embedding AI output into the already complex clinical workflow. Furthermore, in our experience, the environment in which some health care organizations operate often leads

IOMED

C/ de St. Antoni Maria Claret, 167,
Pabellón de La Mercè,
08025 — Barcelona

gabi@iomed.health

https://iomed.health